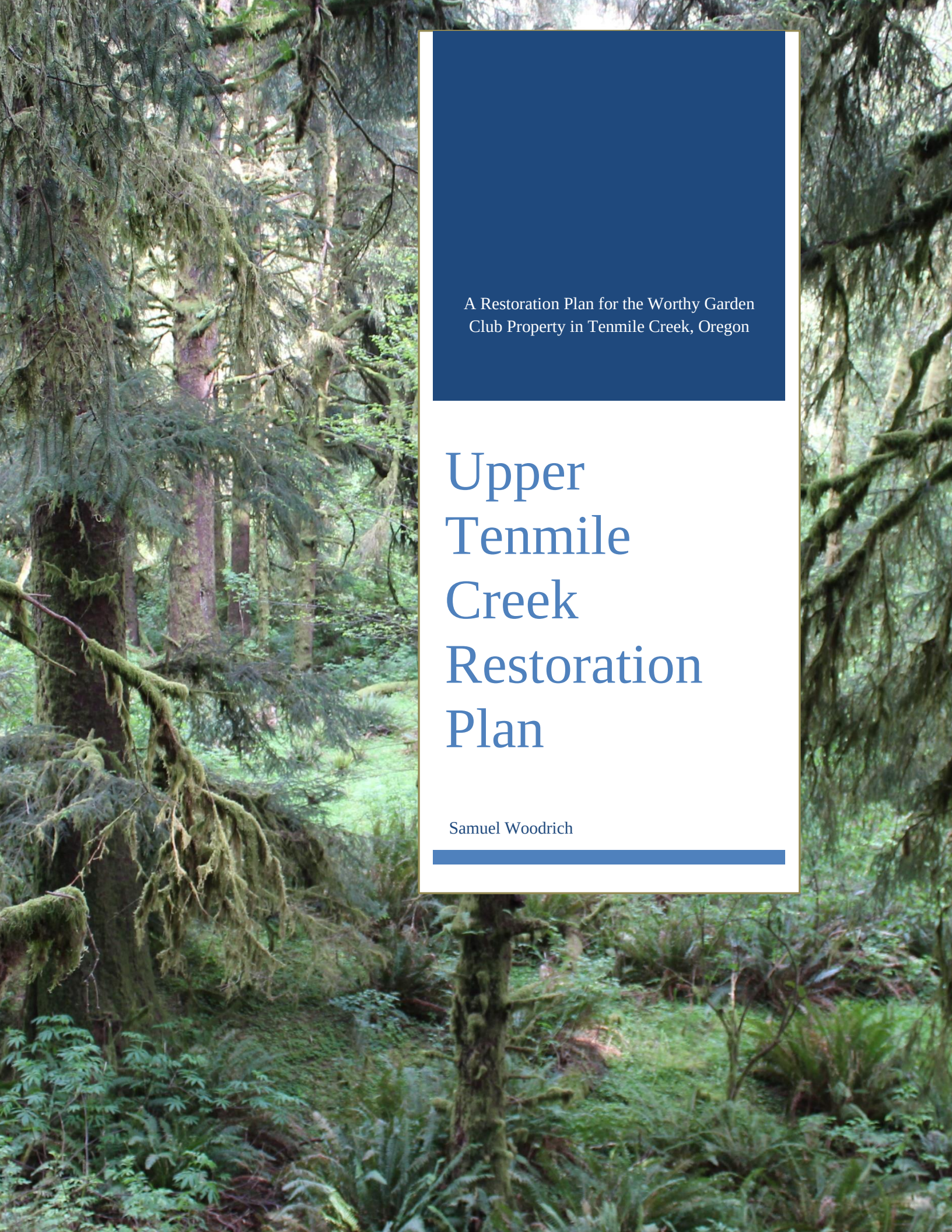

A Restoration Plan for the Worthy Garden Club Property in Tenmile Creek, Oregon

Upper Tenmile Creek Restoration Plan

Samuel Woodrich

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Abstract

The Worthy Garden Club property located in Tenmile Creek, Oregon consists of riparian habitat that is heavily degraded from continuous agricultural use in the 20th century. The functions within the terrestrial and aquatic ecosystems in Tenmile have been altered to various stable states within watershed, although most of the properties that encompass the creek and the riparian areas located in Tenmile exhibit healthier ecological conditions than the Worthy Garden Club property. The Worthy Garden Club property has been designated as a suitable location for ecological restoration strategies that will improve flora diversity, aquatic habitat conditions, soil qualities in riparian areas, aesthetic value of the landscape, and provide future generations with the opportunity to explore coastal watersheds in conditions like those that the indigenous peoples of the confederated Tribes of the Siletz Indians experiences historically. This restoration plan outlines the goal and objectives, costs, monitoring and evaluation strategies, and detailed instructions for the implementation of restoration initiatives meant to improve the ecological, hydrological, and physical conditions of upper Tenmile Creek.

Introduction

Tenmile Creek Oregon is located eleven kilometers south of the central Oregon coastal town of Yachats, Oregon (44.221162, -124.094811). The Tenmile Creek watershed is parceled into several different properties, currently consisting of USFS land, private landowners, timber harvest companies, and restoration projects dispersed within the valley of the watershed. Following the 1906 Forest Homestead Act, immigrants purchased land to establish timber harvesting operations, homesteads for ranching, and private residential properties within Tenmile Creek (Radtke 2017). This transformed the historical landscape from an interconnected ecosystem of riparian areas, wetlands, forests of various successional periods, to a present day fragmented and disturbed watershed. In 2019, the Worthy Garden Club of Bend Oregon purchased a 22.26-hectare (55 acre) plot of land within Tenmile Creek Oregon with intent to restore the site within the coming decades.

Image One: A four bed, one bath house constructed on the eastern part of the property in 1942, converting the property into a private homestead in Lane County, Oregon. The house was demolished in 2019. Photos provided by Worthy Garden Club.

The Worthy Garden Club property is located 10 km east of the Pacific Ocean within the center of the Tenmile Creek watershed. The property line is rectangular in shape, encompasses the creek in its center, and has riparian habitat of varying quality surrounding the creek within the boundaries of the property line **(Appendix Map One and Two)**. The property is relatively flat in topography and consists mostly of land surrounding the creek itself with very little of the higher elevation forests within the valley being within the boundary of the property line. Immediately following the homestead act, the history of early settlement on the property is not certain. It is speculated that the original owners of the property, as well as several other properties in Tenmile Creek was Charles Knowles and Maud M. Miller (Radtke 2017), though official documentation of the legal purchase of the property following the homestead act could not be located. The earliest documentation of official ownership of the property being exchanged occurred in 1930, followed by the first construction of a county registered building on the property in 1942 (Lane County Land Management Division 2019) **(Image One)**. Cattle and sheep ranching on the property was attempted for several decades. However, the isolated location and poor ranching conditions forced the property owners to abandon ranching as a viable way of earning income off the property (Radtke, private conversation). While attempting to ranch on the property, the homesteaders cut down large stands of trees nearby the creek to encourage grass growth and livestock herding, leading to large proportions of properties exemplifying habitat conditions unlike anything documented prior to the enactment of the homestead act. Subsequent anthropocentric effects were witnessed in Tenmile creek, such as the decrease of apex predators that had historically been present in the region, like cougar *(Puma concolor)* and gray wolf *(Canis lupus)*. The direct hunting of these apex predators and the increased presence of humans in Tenmile Creek deterred apex predators from hunting within the region, and the Tenmile valley began to form recruitment gaps in

the canopy due to the increased rate of browsing by native ungulates and livestock grazing in the watershed. Aquatic conditions in the creek began to deteriorate as well, leading to a decline in salmonoid populations, increase in invasive vegetation such as reed canary grass *(Phalaris arundinacea)*, Japanese knotweed *(Polygonum sachalinense*), creeping buttercup *(Ranunculus repens)* and Himalayan blackberry *(Rubus bifrons)*. The replacement of native species with non-native invasives is speculated to have been a contributing factor in widening the creek due to the replacement vegetations' inability to prevent streambank erosion during seasonal flood events (Colleran et al. 2020). Continual timber harvests at the upper elevations of Tenmile Creek may also have contributed to diminishing stream health in the region by increasing sediments present in the stream (May 2002).

Tenmile creek is within the Coast Range level three ecoregion, specifically the Volcanics sub-ecoregion (US EPA 2015) and connected to the Alsea 17100205 watershed (USGS 2019), making it one of the rainiest locations along the central Oregon coast. Seasonal variations in rainfall affect the stream size of Tenmile creek greatly, with stream width being three to five times less than during peak flow during the winter **(Image Two)**. Stream depth can be as deep as 2m during peak flow, followed by shallow or dried up passages within the creek bed present during the dry season. Rainfall averages ~55cm during the winter months and ~5 cm in the summer within the watershed **(Appendix Figure One)** (Radtke, Unpublished data) with temperatures in the summer peaking at ~35 degrees Celsius in recent years **(Appendix Figure Two)** (NOAA 2021). The predominant soil types within the property are Nekoma silt loam, Astoria variant silt loam, Hembre silt loam, Formader loam, and Meda loam series (United States and National Cooperative Soil Survey 2005). The property is located within a known breeding ground for the endangered marbled murrelet *(Brachyramphus marmoratus)* and is also near USFS land where the threatened northern spotted owl *(Strix occidentalis caurina)* has been documented to nest. Coho salmon *(Oncorhynchus kisutch)*, Chinook salmon *(Oncorhynchus tshawytscha)*, and steelhead trout *(Oncorhynchus mykiss irideus)* have historical breeding grounds within the creek but have dwindled in abundance since local colonization of the region.

Image Two: Comparison of water levels during peak flow during the winter (left) and during the summer (right). Stream width and depth decreases as rainfall within the region diminishes during the late spring and summer seasons.

Today, most of the Worthy Garden Club property consists of vegetation associated with degraded landscapes in the Pacific Northwest, along with a few regions within the boundary of the property line that resemble to native flora that was once predominant amongst the landscape. In the degraded portions of the riparian area on the property, native vegetation has little potential to naturally grow due to the warmer temperatures caused by the open fields, lack of canopy cover, lack of soil moisture, and competition with aggressive invasive flora such as reed canary grass. Within the riparian area surrounding

the creek, some pockets of native flora are present, consisting of bigleaf maple *(Acer macrophyllum)*, vine maple *(Acer circinatum)*, red alder *(Alnus rubra*), western hemlock *(Tsuga heterophylla)*, Sitka spruce *(Picea sitchensis)*, red cedar *(Thuja plicata)*, and cascara buckthorn *(Rhamnus purshiana)* as the predominant tree species. Several woody and herbaceous shrubs associated with the central Oregon coastal region are present within these regions of the property, including salmonberry *(Rubus spectabilis)*, red huckleberry *(Vaccinium parvifolium)*, common sword fern *(Polystichum munitum)*, thimbleberry *(Rubus nutkanus)*, red elderberry *(Sambucus racemosa)*, and cow parsnip *(Heracleum maximum).*

There are several other stakeholders that are interested in the restoration initiatives implemented in Tenmile Creek, such as the Midcoast Watersheds Council, Portland Audubon Society, Confederated Tribes of the Siletz Indians, Siuslaw National Forest, and various property owners within the Tenmile Creek watershed. The Midcoast Watersheds Council will be involved in the restoration of Tenmile Creek through their provision of various reports and reviews of restoration projects that have been conducted in various central Oregon coastal watersheds (Garono and Brophy 2001; MidCoast Watersheds Council 2020). Portland Audubon Society and the Confederated Tribes of the Siletz Indians will represent their viewpoints on the restoration project through their provision of salmonid repopulation success data in previously restored regions (MidCoast Watersheds Council 2020), and through their experience in restoring other sections of Tenmile Creek in the past (Bob Sallinger and Paul Engelmeyer 2022). Finally, the Siuslaw National Forest and residents of Tenmile Creek will be represented in the restoration plan through their provision of historical data within the watershed (Radtke 2017) and for their supply of previously collected field data and historical vegetation composition of the region (United States 2005). The actions that are implemented within the restoration site will be guided through the desires and recommendations provided by these stakeholders but will ultimately be decided by the head of the restoration project. Finally, involvement by these stakeholders will be implemented through various meetings with each stakeholder, and through numerous group visits of the property prior to the implementation of any restoration initiatives.

Restoration Goals, Objectives, and Cost

Prior Analysis and Considerations

Most of the riparian area in the Worthy Garden Club property at Tenmile creek had been converted to a livestock pasture consisting of nonnative and invasive grass species as its primary vegetation composition. The anthropogenic degradation caused by livestock ranching has led to an abundance of invasive species that are displacing native forbs and shrubs surrounding the livestock pasture, an increase in surface temperature within the pasture due to the lack of a cooling mesic climate caused by the presence native species, degraded soil characteristics from livestock trampling and lack of canopy cover, altered hydrological conditions caused by the lack of downed woody debris present in the creek and by the current alternative ecological state of invasive vegetation occupying a majority of the banks along the creek, and by the absence of trophic cascade effects that were once caused by apex predator abundance within the watershed.

Following the purchase of the property by the Worthy Garden Club in 2019, some restoration initiatives have taken place starting in 2020 to measure the effect that each initiative would have on the landscape in mitigating these noted ecologically degrading pressures, such as the implementation of

heavy machinery used to mow invasive grass species and blackberry on the site, the felling of trees into the creek to encourage slower streamflow and greater depths within the creek, the placement of large woody debris within the previously grazed pasture, and finally the sowing of native saplings grown in nurseries within the pasture **(Image Three)**. These restoration initiatives have been implemented as experiments testing the resiliency of invasive vegetation to differing severities of removal, hydrological responses to alterations in the flow regimes of the creek, and survival rates of saplings given the current degraded conditions of the habitat for newly planted native flora. This restoration plan integrates the current and future climate conditions on the property and needs to prioritize the mitigation of current herbivory effects on newly planted saplings caused by native ungulates. In the open field where livestock grazed during the 20th century, the soil is greatly compacted, has poor drainage, and retains low amounts of moisture throughout the summer. Roosevelt elk *(Cervus canadensis roosevelti)* have grown fond of the property for its provision of palatable saplings in a watershed that has experienced greater herbivory rates since the decline of apex predator abundance within the region. Furthermore, the wide-open space that the open field provides for elk allows the species to survey the surrounding region for predators with ease. The culmination of these factors has led to a high mortality rate of newly sown saplings within the restoration site, causing a significant financial burden for the Worthy Garden Club, and a minimal impact to improving ecological conditions of the site.

Image Three: In clockwise order starting with the top left image. *A.)* A Sitka spruce *(Picea sitchensis)* sapling planted in the degraded riparian area at Worthy Garden Club. *B.)* Downed woody debris scattered within the degraded riparian area previously used as a grazing pasture for livestock during the 20th century. *C.)* A bobcat excavator with a mower attachment being used to remove Himalayan blackberry *(Rubus bifrons)* from the riparian area on the property. *D.)* A Bigleaf maple *(Acer macrophyllum)* cut down and left as a logjam in Tenmile Creek at the Worthy Garden Club property. All four restoration initiatives were implemented in small scale as part of an experiment measuring their effectiveness as restoration strategies in 2020.

Restoration Goals

The terrestrial preliminary goals of this restoration project will be to remediate the soil on the property, establish native foliage to provide shade and cover within the field that was degraded during the 20th century by livestock ranching, decrease ungulate browsing effects on native species within the property, remove invasive grasses and other flora that outcompete native species, and to create suitable habitat conditions for the passive regeneration of native vegetation from in-tact riparian areas adjacent to the property. The aquatic preliminary goals of this restoration project will be to limit the presence of invasive species along the edges of the creek, reestablish native tree and shrub species to reduce rates of

erosion within the creek bed, create favorable habitat conditions for salmonoid species through the introduction of additional downed woody debris in the creek and greater canopy coverage beside the creek for favorable spawning grounds, and improve general hydrological conditions of the creek itself by creating deeper and narrower channels capable of handling high flow periods during the wet season. The livestock pasture extends from the Tenmile Creek road to the northern part of the creek **(Appendix Map Two)** and serves as a focal point of restoration for this project. Restoration goals on the WGC property in Tenmile Creek consist of satisfying the following objectives.

Restoration Objectives

Ecological

1. Restore hydrological functions in the upper Tenmile Creek watershed while preparing for further degradation outside of the confines of the property.
    a. Increase the populations of seasonal spawning salmonids in the watershed such as Coho salmon *(Oncorhynchus kisutch)*, Chinook salmon *(Oncorhynchus tshawytscha)*, and steelhead trout *(Oncorhynchus mykiss irideus)*.
    b. Reduce creek width and increase depth within the boundary of the property, as well as establishing suitable habitat for native aquatic species.
    c. Reestablish native vegetation along the banks of the creek and decrease seasonal erosion during flooding events.
    d. Maintain restored conditions for several decades through the continual experimentation and studies of ecological conditions in the creek and riparian area, so that uncontrollable alternative climate and hydrological conditions outside of the scope of this restoration plan can be mitigated in the future.

2. Restore the livestock pasture to suitable riparian habitat resembling that of historical riparian conditions while increasing the probability that the restored habitat will remain ecologically functional in increasing drought severity and alternate climate scenarios.
    a. Improve habitat conditions for native species so that passive restoration can occur from surrounding native vegetation patches.
    b. Reintroduce the effects of apex predator abundance on the property so that ungulate herbivory and vegetation/soil trampling decreases in the restoration site.
    c. Increase soil moisture, structure, and fertility.
    d. Reintroduce vegetation that can provide habitat for endangered species during late successional periods of vegetation on the property, such as the marbled murrelet *(Brachyramphus marmoratus)* and the northern spotted owl *(Strix occidentalis caurina)*
    e. Eliminate invasive vegetation species from the property that are capable of displacing native vegetation in current and future climate scenarios.
    f. Monitor the cooling and moisture effects of the newly established mesic conditions to ensure that future climate and drought conditions are mitigated through the establishment of native species and that native species are capable of surviving expected future climate conditions.

Economical

1. Contribute to the ongoing restoration of impaired watersheds within the central Oregon Coast, increasing the ecological functions, and the goods and resources the watersheds provide while improving property values of the surrounding region.
    a. Restore hydrological functions of the creek, reducing the occurrence and fiscal cost flooding incidents within the watershed.
    b. Increase salmonid populations, increasing the funds raised through seasonal and daily fishing licenses within the region.
2. Increase biodiversity and property values within the watershed and decrease wildfire probability of the terrestrial portion of the property.
    a. Increase property values of Tenmile Creek by converting the open pasture patch to a continuous, interconnected riparian area within the watershed.
    b. Decrease potential for wildfires to start and spread within the coastal watershed through the conversion of open grassland pastures to a damp mesic climate that is less likely to supply fuel for a wildfire.

Cultural

1. Support the re-establishment of seasonal salmon runs in Tenmile Creek that are similar in size to historical numbers.
    a. Increase the probability of future generations being able to witness salmonid spawning in the central Oregon coast by increasing the quality of habitat for salmon in Tenmile Creek
    b. Serve as a model for successful restoration plans in the central Oregon coast and present the opportunity for future research to be conducted within the watershed.
    c. Increase populations of culturally important species that can serve as a source population for downstream habitats.
2. Improve the aesthetic value of the restoration site by converting the decommissioned livestock pasture into riparian habitat resembling previous habitat conditions before the establishment of the Forest Homestead Act of 1906.
    a. Provide future generations with a natural experience in a field setting that resembles the conditions previous native Americans and early colonists encountered prior to the conversion of riparian habitat for agricultural or silvicultural purposes.
    b. Establish suitable habitat for endangered species so that future generations are not exposed to diminishing biodiversity rates in coastal ecosystems.

To satisfy all the ecological goals and objectives of the restoration project mentioned, several restoration initiatives will need to be implemented in both the terrestrial and aquatic habitats of the Worthy Garden Club property in Tenmile creek **(Table One & Two)**. To achieve all previously mentioned goals for this restoration project, all initiatives will need to be implemented in the *Complete Restoration* plan. However, there are two additional alternatives that are considered for the site if there are economic constraints or temporal limitations for the project. These alternative methods to the *Complete Restoration* plan are deemed, *Partial Restoration Alternative* and *No Action Restoration Alternative* and follow similar trajectories for restoration as the *Complete Restoration* plan but are reduced in the number of initiatives implemented on the property.

The aquatic initiatives listed involve altering the hydrology of the restoration site and will require a removal-fill permit granted by the state of Oregon based on Oregon's Removal-Fill Law (ORS 196.795-

900). Siuslaw National Forest will also be involved in planning the alteration of Tenmile Creek's hydrology since a majority of the watershed is located within publicly owned land and managed by the United State Forest Service, though an official permit for hydrological restoration of the site is not necessary since the focal point for management is within boundary of the property owned by Worthy Garden Club. The Oregon Department of Fish and Wildlife will also be consulted for the agency's expertise and management of stream corridors passable by native fish species. Finally, all restoration initiatives will need to be reviewed by a land manager of Lane County since the property is located within an impacted forest land zone (F-2).

Terrestrial Initiatives	***C***	***P***	***N/A***
Mowing of non-native and invasive grasses present in decommissioned livestock pasture.	**X**	**X**	**X**
Introduction of woodchip mulch onto the soil surface of the pasture for improved soil aggregate structure, moisture content, and fertility.	**X**	**X**	
Sowing of native tree saplings and shrub species in the decommissioned livestock pasture.	**X**		
Play recorded apex predator sounds from various speakers placed within the restoration site on regular intervals to decrease ungulate browsing potential of native saplings and plants.	**X**	**X**	**X**
Semi-regular watering of sown native saplings in the decommissioned pasture to prevent high mortality rates in the summer.	**X**		
Collect several metrics of habitat quality following the implementation of restoration initiatives such as climate and temperature, soil moisture and infiltration rate, biomass and diversity of vegetation, and herbivory/browsing data.	**X**	**X**	**X**

Aquatic Initiatives	***C***	***P***	***N/A***
Downed woody debris is introduced into the creek by felling red alder *(Alnus rubra*) on the banks of the creek into the stream.	**X**		**X**
Hand removal of reed canary grass *(Phalaris arundinacea)* and Japanese knotweed *(Polygonum sachalinense)* along the banks of the creek.	**X**		
Supplemental planting of bank stabilizing and erosion preventing plant species such as Pacific willow *(Salix lasiandra)*, bigleaf maple *(Acer macrophyllum)*, western hemlock *(Tsuga heterophylla)*, and salmonberry *(Rubus spectabilis)*.	**X**		
Ecological monitoring of the stream temperature, width and depth, and salmonid populations will be conducted following the implementation of restoration initiatives.	**X**		**X**

Table One and Two: Terrestrial and aquatic restoration initiatives for the Worthy Garden Club restoration project in Tenmile Creek, Oregon. The implementation of each restoration initiative is indicated by having an "**X**" within the columns for the *complete restoration* plan (**C**), *partial restoration alternative* (**P**) and *no action restoration alternative* (**N/A**) plans respectively.

Complete Restoration Plan

The complete restoration plan will meet the ecological goals of the Worthy Garden Club restoration site by implementing several terrestrial and aquatic restoration initiatives on the property. Aquatic restoration in the site will be achieved by felling red alder *(Alnus rubra*) that is located along the streambank into the creek. The addition of downed woody debris into the creek will provide suitable habitat for aquatic species and change the creek morphology to resemble hydrological conditions that were previously witnessed in Tenmile Creek prior to the colonization of the watershed. Invasive species such as reed canary grass *(Phalaris arundinacea)* and Japanese knotweed *(Polygonum sachalinense*) will be removed by hand along the banks of the creek and replaced with bank stabilizing and erosion

preventing native plant species, such as Pacific willow *(Salix lasiandra)*, bigleaf maple *(Acer macrophyllum)*, western hemlock *(Tsuga heterophylla)*, and salmonberry *(Rubus spectabilis)*. Following the establishment of these restoration initiatives, continued monitoring of the hydrological conditions of the creek will be conducted, including the measurement of the creek temperature, width and depth during peak flow, and salmonid population monitoring during spawning season.

Terrestrial restoration will involve the mowing of non-native and invasive grass species present in the decommissioned livestock pasture. Following this initiative, woodchip mulch will be imported to the site and distributed throughout the mowed pasture, allowing nutrients to be reintroduced into the soil, improving soil aggregate structure, moisture content, and fertility. Native tree saplings and shrub species will be planted throughout the decommissioned pasture so that a mesic climate can be established as successional periods of the landscape progress. Semi-regular watering of newly sown saplings will be implemented in the summer months so that the probability of sapling survival is greater in the early years of vegetation establishment. To prevent native ungulates from browsing the newly planted saplings, or the native plants that immigrate from source populations surrounding the pasture, speakers will be placed around the perimeter of the pasture that regularly broadcast recorded sounds of apex predators native to the region, such as the cougar *(Puma concolor)* and gray wolf *(Canis lupus).* Finally, terrestrial habitat quality will be monitored following these restoration initiatives, such as regular climate and temperature monitoring in the restored riparian area, soil moisture and infiltration rate tests, the collection of biomass and plant diversity indices, and browsing/herbivory monitoring of native species.

Partial Restoration Alternative

The partial restoration alternative plan involves the implementation of some initiatives mentioned in the complete restoration plan, such as the mowing of non-native and invasive grass species, introduction of woodchip mulch to improve soil conditions, implementation of apex predator audio recordings played through speakers within the restoration site, and the collection of data following the implementation of these terrestrial restoration initiatives. Additionally, aquatic restoration initiatives consist of the felling of red alder into the creek so that stream morphology is changed, the hand removal of invasive species, and the monitoring of stream characteristics following the implementation of these initiatives. The partial restoration plan is designed for the rehabilitation of landscape characteristics during the initial implementation of the restoration plan followed by a reliance on semi-passive restoration initiatives to improve physical and ecological conditions of the site.

No Action Restoration Alternative

The no action restoration alternative plan consists of minimum anthropocentric alteration of the landscape and relies heavily on the passive restoration of the site though source populations of flora and fauna from surrounding in-tact riparian habitats. This includes the mowing of non-native and invasive grass species, and the implementation regularly broadcasted apex predator recordings on the terrestrial region of the restoration site, and the felling of red alder into the creek so that stream morphology is altered. However, like the other alternatives to this restoration plan, routine experiments and studies involving the assessment of terrestrial and aquatic habitat quality will be conducted following the implementation of these initiatives. The no action restoration plan would mitigate the primary drivers for degradation in the aquatic and terrestrial habitats at the Worthy Garden Club property while allowing for the passive reestablishment of historical hydrological regimes and for the natural reintroduction of native vegetation and historical soil conditions on the property.

Preferred Alternative – Complete Restoration Plan

The complete restoration plan drafted for the Worthy Garden Club property at Tenmile Creek was selected as the preferred alternative state for the site since it addresses the most issues related to the current hydrological and terrestrial conditions of the restoration site. As part of the criteria for successful restoration within riparian environments (Palmer et al. 2005), the complete restoration plan captures a suitable endpoint for the project, outlines how ecological conditions of the site are improved following restoration, suggests methods for increasing resilience, avoids the implementation of restoration initiatives that harm the landscape and ecological functions to a more severe degree than what is present before restoration work is conducted, and finally incorporates post restoration project monitoring. Each of the proposed actions for the complete restoration plan will be implemented in sequential order from top to bottom in the cost analysis provided below **(Table Three & Four)**. Terrestrial and aquatic initiatives will first be implemented in tandem with one another and will be monitored after the restoration plan has been completed in Tenmile Creek. Subsequent monitoring of the impact of the restoration plan will be conducted in the summer after restoration work is completed and again ten years following the completion of the restoration plan by graduate students in the Environmental Sciences Graduate Program at Oregon State University.

The implementation of the following restoration initiatives will be conducted over the course of three years, followed by monitoring for an additional ten years after completion of the restoration project. Most initiatives are affiliated with temporal restraints, such as the sowing of native saplings and tree species on the property and will require additional planning revolving around suitable environmental conditions on the site and collaboration with greenhouses of native species. Some initiatives are speculative and may not need to be implemented during the plan, such as the supplemental watering of planted saplings if weather conditions are favorable during the summer or as environmental conditions on the landscape improve over time, such as soil fertility and moisture content increasing after mulching the property. All costs are predicted based on current market conditions and rates for restoration work in the Central Oregon Coastal region and are subject to change. The grand total for the implementation of the complete restoration plan at the Worthy Garden Club property in Tenmile Creek is $98,455. Terrestrial restoration initiatives will be implemented within the 6.20-acre, decommissioned livestock pasture on the property and aquatic restoration initiatives will be implemented along the north and south sides of Tenmile creek adjacent to the pasture.

Terrestrial Initiatives	***Cost***	***Hours***	***Unit***	***Total***
Mowing of non-native and invasive grasses present in decommissioned livestock pasture.	**$925/hour**	**10**	**-**	**$9,250**
Introduction of woodchip mulch onto the soil surface of the pasture for improved soil aggregate structure, moisture content, and fertility.	**$6,275/acre**	**-**	**6.20 acres**	**$38,905**
Sowing of native tree saplings and shrub species in the decommissioned livestock pasture. Sapling and shrub species will be determined through the collaboration between the restoration project manager and greenhouses or native plant growing operations	**$5/sapling and shrub**	**-**	**500 plantings**	**$2,500**
Play recorded apex predator sounds from various speakers placed within the restoration site on regular intervals to decrease ungulate browsing potential of native saplings and plants.	**$800/broadcasting equipment**	**-**	**2 sets needed**	**$1,600**
Semi-regular watering of sown native saplings in the decommissioned pasture to prevent high mortality rates in the summer.	**$750/watering**	**-**	**Estimated 10 instances**	**$7,500**
Collect several metrics of habitat quality following the implementation of restoration initiatives such as climate and temperature, soil moisture and infiltration rate, biomass and diversity of vegetation, and herbivory/browsing data.	**$4,000/Field season**	**-**	**3 Field seasons**	**$12,000**

Aquatic Initiatives	***Cost***	***Hours***	***Unit***	***Total***
Downed woody debris is introduced into the creek by felling red alder *(Alnus rubra*) on the banks of the creek into the stream.	**$1,250/hour**	**10**	**-**	**$12,500**
Hand removal of reed canary grass *(Phalaris arundinacea)* and Japanese knotweed *(Polygonum sachalinense)* along the banks of the creek.	**$120/hour**	**10**	**-**	**$1,200**
Supplemental planting of bank stabilizing and erosion preventing plant species such as Pacific willow *(Salix lasiandra)*, bigleaf maple *(Acer macrophyllum)*, western hemlock *(Tsuga heterophylla)*, and salmonberry *(Rubus spectabilis)*.	**$5/sapling**	**-**	**200 plantings**	**$1,000**
Ecological monitoring of the stream temperature, width and depth, and salmonid populations will be conducted following the implementation of restoration initiatives.	**$4,000/Field season**	**-**	**3 Field seasons**	**$12,000**

Table Three and Four: Terrestrial and aquatic restoration initiatives for the Worthy Garden Club restoration project in Tenmile Creek, Oregon. Restoration initiatives are ordered from top to bottom by time of implementation, with the first row being the initial restoration initiative implemented on the project. Grand total of the restoration project is $98,455 and will be implemented within three years of the starting date for the restoration project.

Reference System Comparison, Restoration Monitoring, and Outreach

The success of the Tenmile Creek restoration project at the Worthy Garden Club property will be evaluated in comparison to a reference riparian area located further downstream within the same watershed along the central Oregon coast. Eight kilometers west of the restoration project site is a parcel of land managed by Siuslaw National Forest Service that has remained unaltered since its acquisition during the late 1800s. Due to the regions lack of accessibility for loggers compared to higher elevation regions within the watershed, timber harvest operations were not conducted within the riparian area of

lower Tenmile Creek (Geospatial Enterprise Office 2023), resulting in a late successional canopy composition of vegetation unlike other parcels of land within the watershed. Furthermore, high severity fires within the reference site are estimated to not have occurred within the past 200 years, whereas most of the land within the watershed, including the restoration site located in Tenmile Creek were disturbed by the high severity wildfire such as the 19th century coastal wildfire (Northwest Oregon Area Ecology Program 2022). Although the reference site for the restoration project primarily consists of a late successional composition of vegetation within the region, restoration initiatives implemented at the Worthy Garden Club property have been selected based on the current conditions of the reference site. The reference site serves as a desirable state for the restoration site to strive to achieve, and as a template for comparison of the terrestrial and hydrological conditions present within the watershed after restoration has taken place. The biotic community composition of the reference site has been collected as part of an ecological study being conducted by the Oregon State Environmental Sciences Department and will be reviewed in this restoration plan to determine the desirable vegetation community composition for the restoration site after all initiatives have been implemented. Additionally, information regarding the physical environment of the reference site will need to be collected before hydrological restoration initiatives are implemented within the restoration site.

Reference Site Biotic Community Composition

As part of a larger ecological study conducted in Tenmile Creek, predominant vegetation within the watershed was collected in 2022, serving as an adequate cataloging of herbs, shrub, sapling, and tree species composition within the reference site. In doing so, occurrences of invasive species within the reference site were quantified, granting additional insight to the niches that are fulfilled by invasive species in a healthy riparian habitat in the watershed.

Herbs and shrubs with the highest number of encounters by species were found to be mostly native species, with the native western sword fern (*Polystichum munitum)* being the most encountered species followed by the introduced and invasive Japanese knotweed *(Polygonum sachalinense)* being the second most encountered species in the reference site (Woodrich, Unpublished Data) **(Figure One).**

The results of the recorded dominance study revealed that the most predominant species found within the reference site were native. Additionally, native species were found to have a greater richness in comparison to invasive species of vegetation. However, while invasive species had less overall richness, invasive species in the reference site were somewhat prominent amongst all shrub and herb species recorded (Woodrich, Unpublished Data) **(Figure Two).**

Finally, all tree species sampled in the reference site found to be native, with an increased number of trees found compared to saplings. Saplings of Sitka spruce *(Picea sitchensis)* and western hemlock *(Tsuga heterophylla)* were found in greater abundance compared to adult trees of the same species while red alder *(Alnus rubra)* and bigleaf maple *(Acer macrophyllum)* were found in greater abundance as trees than saplings (Woodrich, Unpublished Data) **(Figure Three).**

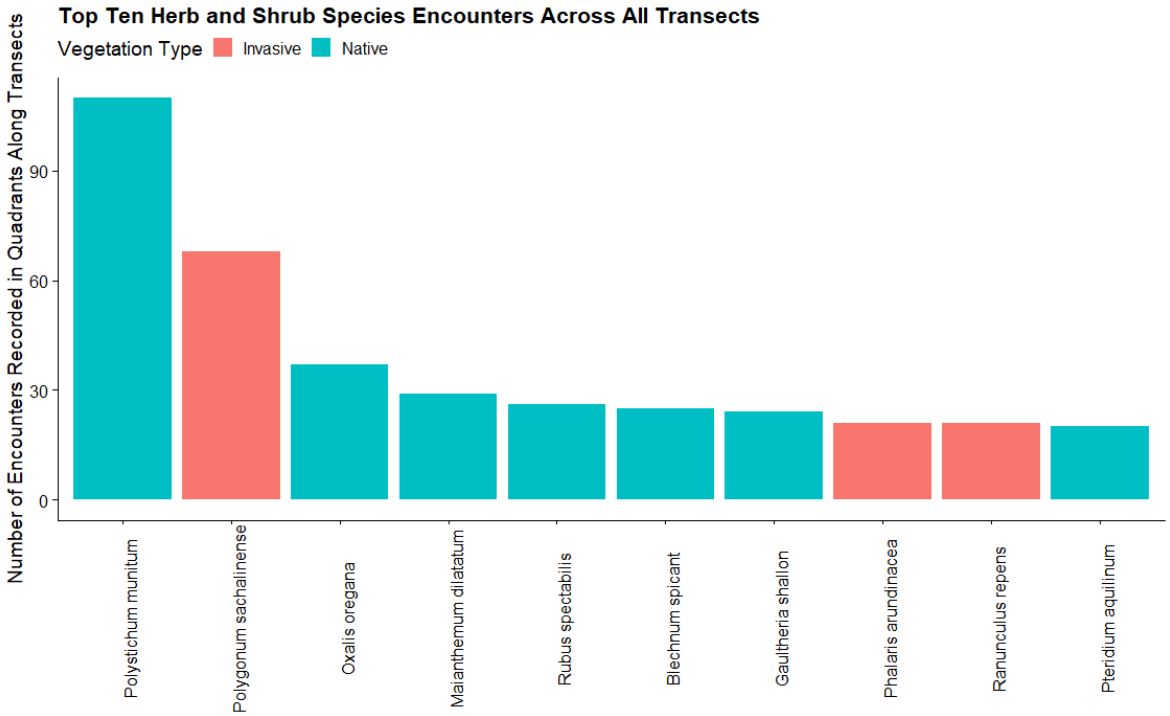


Figure One: The top ten most encountered shrub and herb species found within the reference site of Tenmile Creek. Higher numbers of encounters represent a greater number of times a species was present within the reference site.

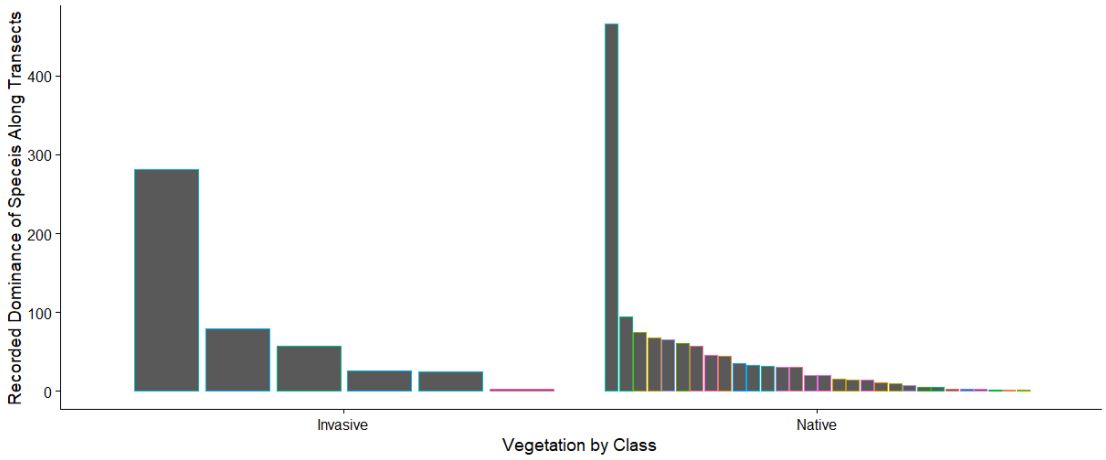


Figure Two: Vegetation species dominance within the reference site of Tenmile Creek. The recorded dominance of a species represents the sum of each species assigned predominance value determined by the cover amount of the species in each quadrant across all transects used in the sampling of vegetation. A greater recorded dominance value indicates that a species of vegetation had a greater cover percentage within the reference site. A lower recorded dominance value indicates that a species of vegetation had a lower cover percentage within the reference site.

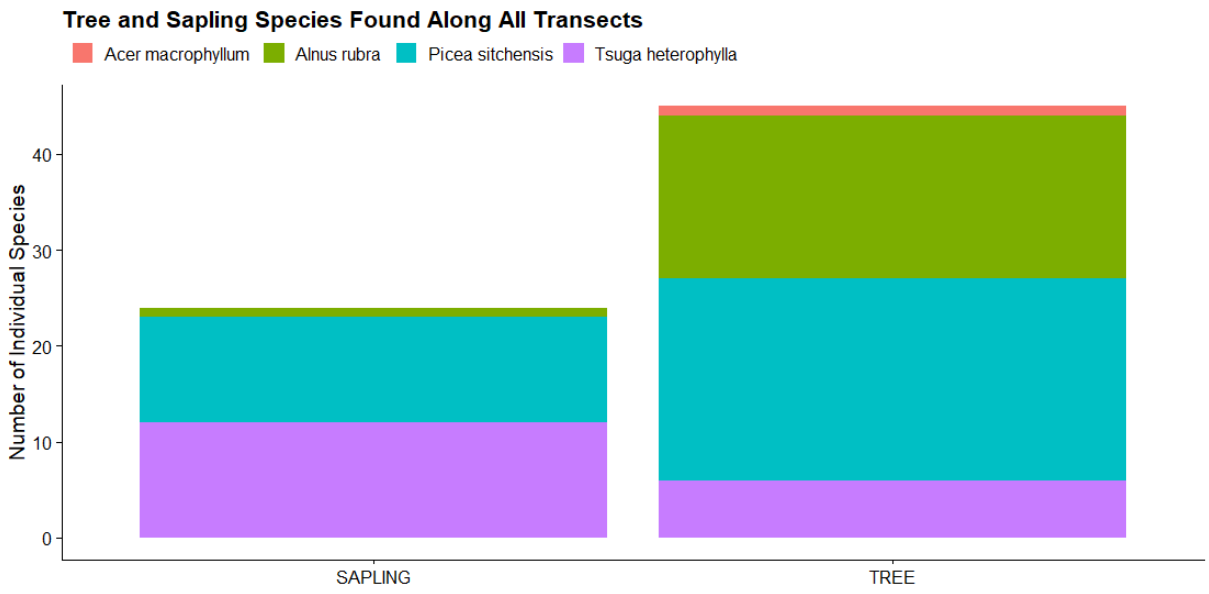


Figure Three: Number of tree and sapling individuals sampled by species within the reference site of Tenmile Creek, Oregon.

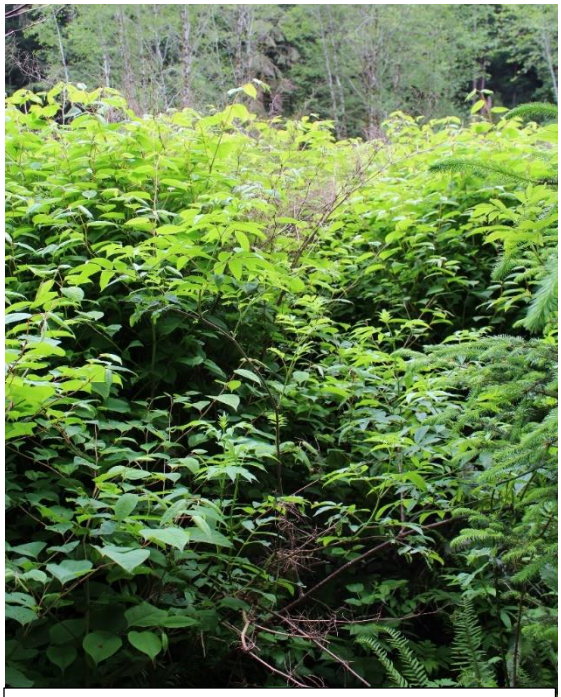


Image Four: Japanese knotweed (*Polygonum sachalinense)* patch located near the bank of the reference site in Tenmile Creek along a transect with red alder *(Alnus rubra*) in the background.

Six species (17%) of vegetation encountered in the sampling of the reference site were found to be invasive species, while the thirty other most encountered species were native, indicating that the native vegetation can be naturally resilient to invasive species within the reference site in Tenmile creek. Furthermore, invasive species immigration is presumed to be greater in the reference site due to its location being further downstream from the restoration site within the watershed, causing invasive species to be transported there in greater concentrations from private properties and timer operations within the watershed. The immediate banks adjacent to the creek were found to be one of the more degraded regions of the reference site. The creek sides consisted of 5m thickets of Japanese knotweed *(Polygonum sachalinense)* on either bankside that expanded from the creek outward to the riparian area of the site **(Image Four)**. However, past the immediate proximity of the creek, the riparian area of the reference site was composed of predominantly native vegetation, indicating a greater resilience to invasive species outside of the regions where flooding disturbance regularly occurs in Tenmile Creek **(Image Five)**. Finally, tree species composition data collected in

Image Five: Example of vegetation composition within the riparian area of the reference site in Tenmile Creek. Sedge species *(Carex spp.)*, western sword fern *(*Polystichum munitum*)*, and a Sitka spruce *(Picea sitchensis)* are present in the foreground, followed by Japanese knotweed (*Polygonum sachalinense)* in the background, closer to Tenmile Creek.

the reference site indicated a predominant presence of late successional conifer species such as western hemlock *(Tsuga heterophylla)* and Sitka spruce *(Picea sitchensis)* as well as early successional deciduous species like bigleaf maple *(Acer macrophyllum)*, and red alder *(Alnus rubra*). Early successional species were most predominantly found within the immediate area of the creek where invasive species had not become established. Restoration initiatives in the restoration site will mimic this vegetation establishment pattern through the removal of invasive species present and through the assisted planting of early successional deciduous species along the banks in the restoration site.

Reference Site Physical Characteristics

The aquatic conditions of the reference site will need to be collected before implementing any hydrological restoration initiatives on the Worthy Garden Club property. These conditions include stream width, depth, and discharge rates within the system, presence, or absence of downed woody debris within the creek, and canopy cover of native vegetation surrounding the creek. All hydrological metrics will be compared to the current conditions of the restoration site and will be used as a measurement of the degree to which successful aquatic restoration conditions have been implemented following the implementation of the Worthy Garden Club restoration project. Although historical flooding disturbances within the creek have not been captured, collecting information pertaining to the high-water marks of the creek, area of floodplain impacted by flooding disturbances, and soil characteristics may prove beneficial in predicting the severity of flooding possible within the restoration site following the implementation of aquatic restoration initiatives.

Monitoring Success of Restoration Initiatives

Terrestrial based restoration initiatives will be monitored through the multiple temporal comparisons made before, during and after the restoration project in addition to spatial comparisons drawn between the reference site and the restoration project site. Pre-restoration conditions related to the complete restoration project objectives such as soil conditions, native flora diversity, and ungulate presence will be monitored through soil core collections, vegetation sampling through the use of quadrat and transect methods that have been standardized in restoration work (Dybala et al. 2019) and through the survival rates of a subsample of saplings planted within the restoration site, and game camera placement within the restoration site. Hydrological initiatives will be monitored through stream conditions, such as creek widths, depths, and temperature, samples during quarterly portions of the year, capturing erosion rates along the sides of the creek following the establishment of native vegetation each year, and through salmonid counts during spawning season. These monitoring practices will be conducted by graduate students in the Environmental Sciences department as part of an internship program provided by the

school for professional science master's students. One of the three field seasons will take place one calendar year after the implementation of the last initiative in the complete restoration plan, another three years after completion, and finally again ten years after completion. Finally, the monitoring data collected at the restoration site will also be compared to the single year data collected within the reference site and will not be recollected during each established field season.

Successful implementation of the complete restoration plan consists of the following criteria, though the criteria for success may shift as information from each monitoring period is collected. Carbon, nitrogen, and sulfur concentrations and soil moisture/infiltration rate within the decommissioned livestock pasture will improve in percentage per soil core sample each year samples are taken. Native plant diversity will improve each year until the maximum number of niches available for native flora to obtain each year begins to diminish in growth. The occupancy of non-native, invasive flora does not grow between the years that restoration monitoring is conducted. Native ungulate presence within the site decreases following the implementation of apex predator audio broadcasts and the subsample of saplings planted has an average survival rate of 75% since initial planting. Erosion rates of the streambanks do not increase more than 5% and remain the same since the implementation of native saplings along the banks of the creek. Stream temperatures decrease, depth increases, and width decreases since the implementation of aquatic restoration initiatives on the site, and salmonid abundance increases each spawning year that monitoring data is collected.

Public Outreach and Information

Public outreach describing the purpose and efforts involved with the restoration work at the Worthy Garden Club property in Tenmile Creek will involve local outreach programs with the residents of Tenmile and involvement in community meetings held by the Midcoast Watershed Council. Since the population on Tenmile Creek is below 100, signs portraying the restoration work being conducted at the property will be constructed. Additionally, pamphlets will be available attached to the signs that address future and current restoration strategies, their effects on the physical and ecological conditions of the site, and a statement involving the rationale behind the importance of restoring riparian habitats along the central Oregon coast. Community meetings held by the Midcoast watershed council will be attended quarterly in Newport, Oregon. Each meeting grants the opportunity to hear the opinions of and collaborate with other restoration companies, native tribes, and residents of the Oregon coast. Each meeting, a summary of the progress made at the restoration site, images before and after restoration initiatives have been implemented, and future goals and directions for the restoration plan will be shared. Finally, volunteer opportunities will be broadcast at these events, resulting in additional labor for restoration work conducted on the Worthy Garden Club property.

Conclusion

This restoration proposal integrates the opinions of various stakeholders involved with coastal riparian watersheds along the central Oregon coast and portrays a series of criteria outlining the definition of a successful restoration project, and the methodology necessary to achieve these objectives. Through various terrestrial and aquatic remediation strategies, the riparian area of the Worthy Garden Club property can be improved ecologically, physically, and aesthetically for future generations to enjoy, water quality and native species habitat to improve, and for the protection and restoration of a coastal ecosystem that has been historically threatened with environmental degrading pressures caused by the displacement

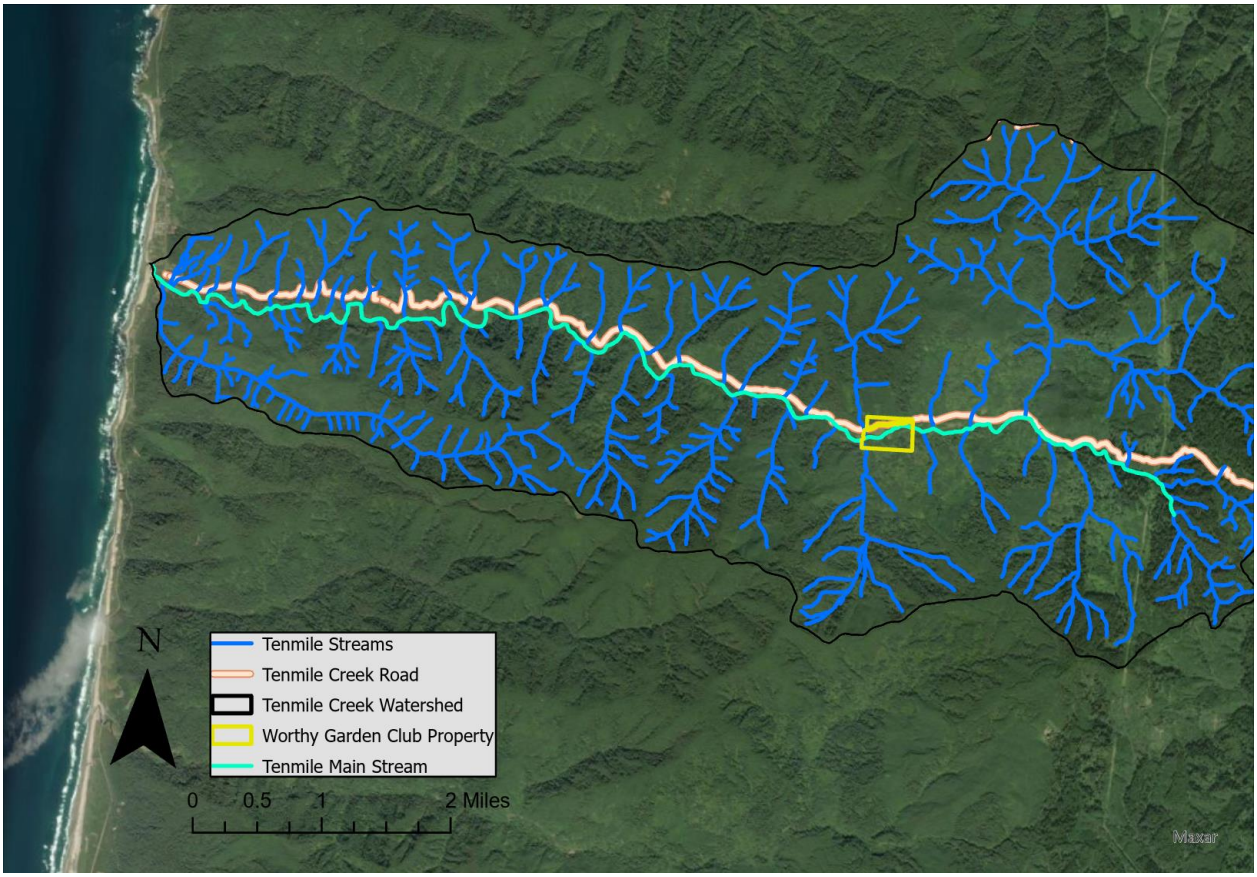
of native species, trophic shifts in the regional food web, increasing disturbance rates and severities, and through the repurposing of landscapes for agricultural and silvicultural use.

Week 10 Reading and Synthesis Addendum

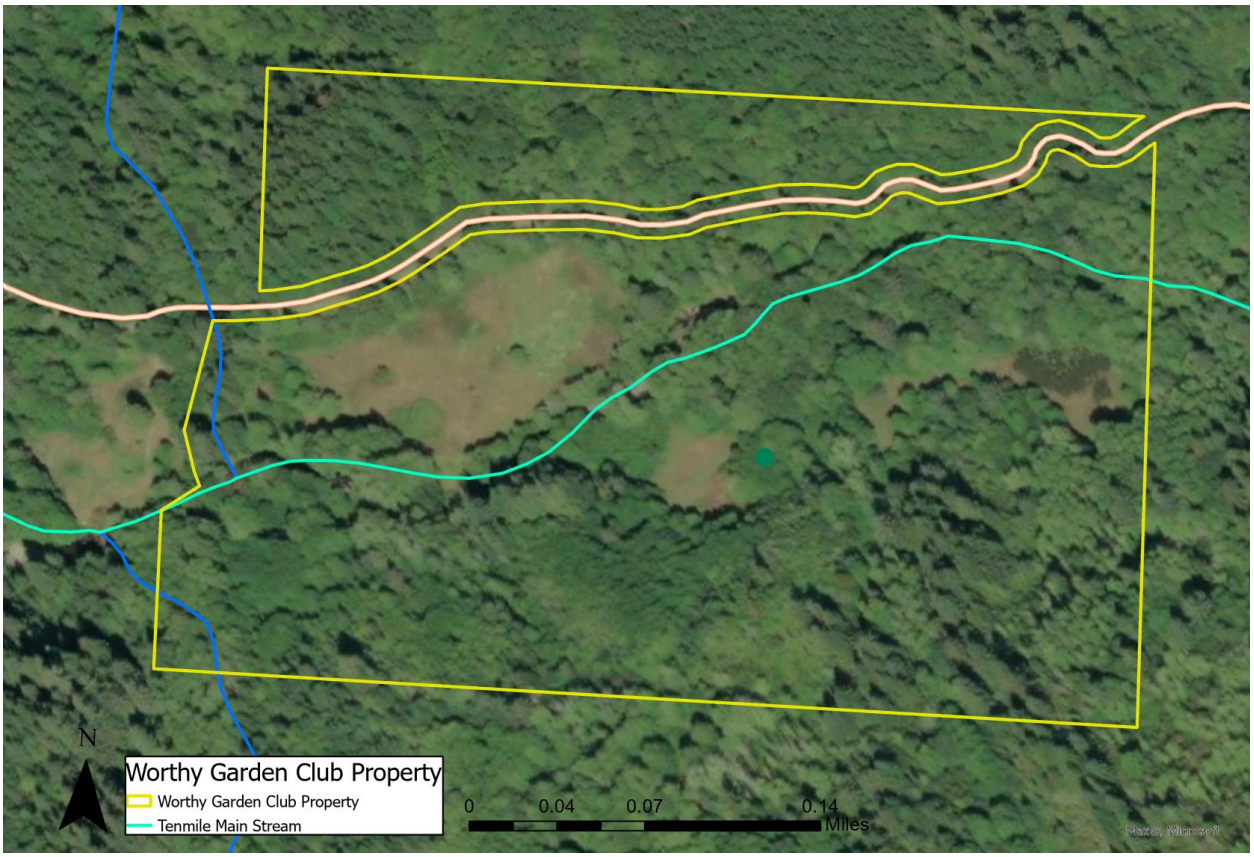
The restoration objectives and goals outlined in this plan are based upon the multiple equilibrium state ecology theory (Donald A. Falk et al. 2006), since the temporal scale involved with the Worthy Garden Club restoration site is large and exhibits a current state of ecological equilibrium that is far from what is perceived as a natural state within the watershed. While the spatial scale of the stie is small, the ecological processes that are present within the restoration site have diverged greatly in comparison to the reference site, such as seasonal flooding severity in riparian zones, native vegetation presence and their impact on erosion rates and bank stabilization, ungulate presence and the trophic effects they have on vegetation composition, mesic climate conditions, and soil characteristics such as fertility, nutrient availability, water holding capacity, and infiltration rates. Some of these processes cannot be fully mitigated, since their effects are felt at a greater spatial region than that of the restoration site, and can only be temporarily mitigated for, such as the implementation of apex predator audio broadcasts at the restoration site. These larger processes will continue to destabilize any current state of the restoration project, pushing the point of ecological equilibrium into another state. In the absence of the mitigation strategies implemented in this restoration plan, these alternative stable states may be closer towards the degraded conditions the restoration site is in prior to the implementation of restoration initiatives. On the contrary, the culmination of fast processes may push the state of the restoration site towards a healthier condition that more closely resembles that of the reference site, such as the combined pressures caused by native vegetation establishment both passively and actively within the decommissioned livestock pasture improving soil conditions, or the reestablishment of native vegetation and introduction of downed woody debris providing suitable habitat for aquatic species, improving biodiversity rates within the riparian area.

Regardless of the direction that these initiatives will guide the ecological state of the restoration site, there is no conceivable final point of equilibrium that the site will ultimately meet. This is because the restoration initiatives depicted in this plan are addressing the ecological issues that are apparent and important to stakeholders currently while drawing comparisons to the historical conditions of the site and current conditions of reference sites, all of which are in or will continue to shift in points of equilibrium throughout time. This is not meant to discourage restoration from taking place but should be recognized and incorporated into the restoration process as the landscape begins to shift and ecological process change following the implementation of each initiative and the environmental pressures that present themselves in the future. The recognition of these shifting baselines for restoration efforts must be incorporated into restoration plans of the central Oregon coast so that ecological improvements of these systems can not only be implemented, but also recognized by those who are implementing them and those who benefit from restoration ecology.

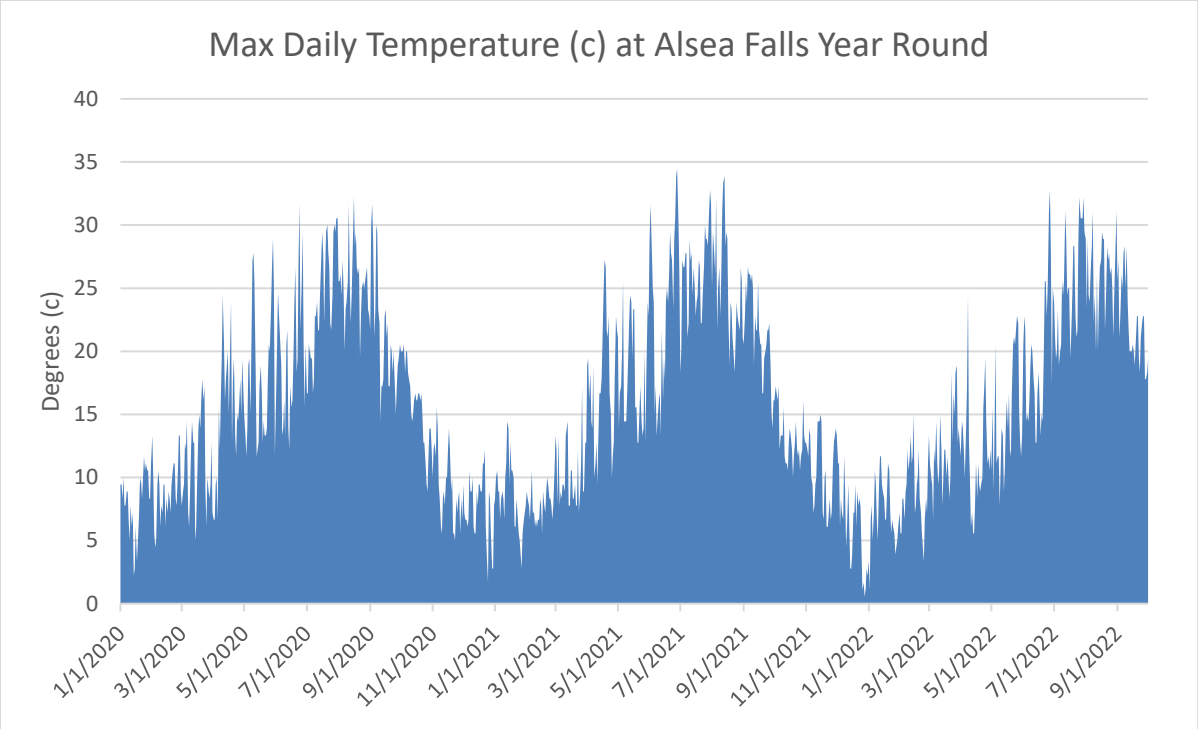
Appendix



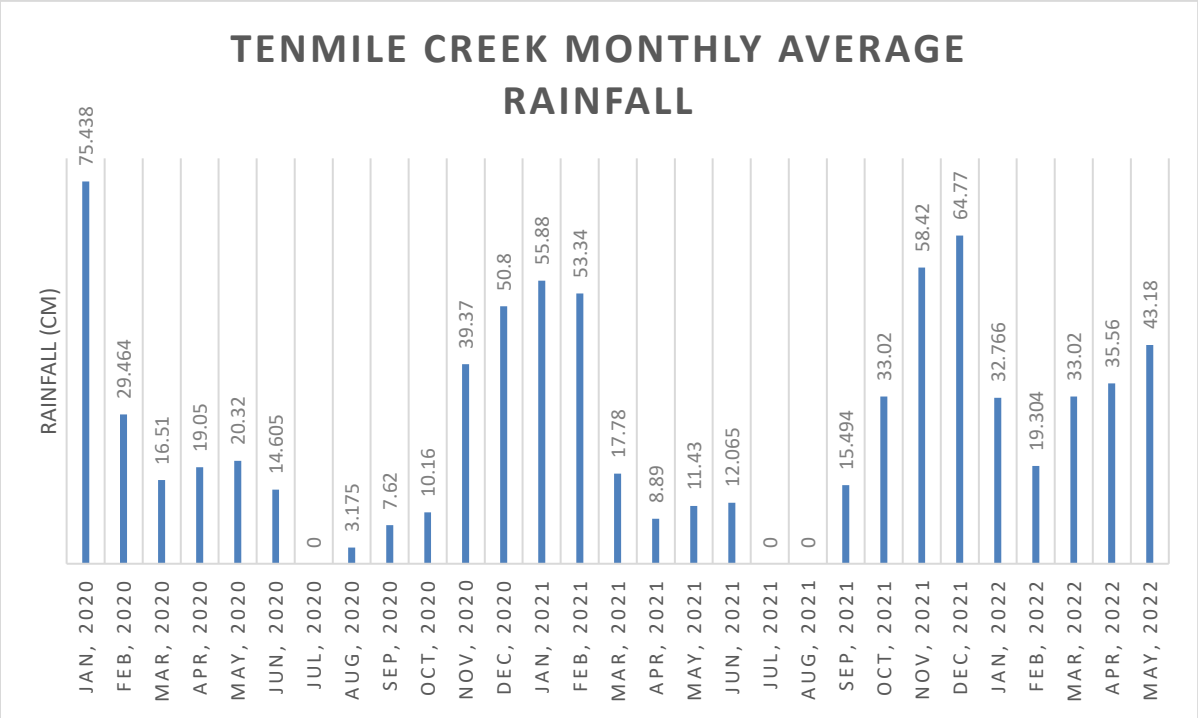
Appendix Map One: Tenmile Creek watershed located 11km south of Yachats Oregon along the central Oregon coast. The 55-acre Worthy Garden Club property is highlighted in yellow 10km east of the Pacific Ocean.



Appendix Map Two: Worthy Garden Club property line encompassing Tenmile Creek. Open patches in the canopy that are brown in color signify areas within the property that were used for ranching and are now considered degraded riparian habitat.



Appendix Figure One: Monthly average rainfall records at Tenmile Creek, Oregon between the years 2020 and May of 2022. Data collected by Hans Radtke, a local resident of Tenmile Creek, Oregon.



Appendix Figure Two: Daily max temperature records at Alsea Falls Creek weather station provided by NOAA – National Centers for Environmental Information between the years 2020 to 2022.

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